

Industrial Controls Release Assurance Package

Project: Release Assurance — DC1_MEP_PLC01

Controller: DC1_MEP_PLC01

Baseline: baseline.L5X

Revised: revised.L5X

Governing SOO/FDS: soo.pdf

Exported: 2026-07-22T16:53:42.821Z

Reviewer: Lead Controls Engineer (reviewer)

NOTICE: This package is a human-reviewed, read-only audit aid. It does not approve a release. A qualified controls engineer must explicitly accept or reject each finding.

Review Summary

Total	Pending	Accepted	Rejected	Edited	Unresolved
16	14	1	0	0	1

Change / Impact Register

ID	Status	Severity	Class	Location	Requirement	Test ID
REV-001	Accepted	Medium	Sequences	P_DC_Cooling/R30_CHWP_Control/rung:2	MAP-DIFF-001	TC-F2S-P102
REV-002	Pending	Medium	Sequences	P_DC_Cooling/R30_CHWP_Control/rung:3	MAP-DIFF-002	TC-F2S-P201
REV-003	Pending	Critical	Interlocks	P_DC_Cooling/R30_CHWP_Control/rung:5	REQ-013	TC-F2S-P102
REV-004	Unresolved	Medium	Alarms/Timers	P_DC_Cooling/R30_CHWP_Control/rung:9	MAP-DIFF-004	TC-F2S-P201
REV-005	Pending	Critical	Interlocks	P_DC_Cooling/R40_AHU_Control/rung:3	REQ-001	TC-F2S-P201
REV-006	Pending	High	Modes	P_DC_Cooling/R40_AHU_Control/rung:5	MAP-DIFF-006	TC-F2S-P201
REV-007	Pending	Medium	Alarms/Timers	P_DC_Cooling/R40_AHU_Control/rung:6	REQ-013	TC-F2S-P101
REV-008	Pending	Medium	Alarms/Timers	P_DC_Cooling/R50_Alarms/rung:13	REQ-016	TC-F2S-P201
REV-009	Pending	Medium	Alarms/Timers	P_DC_Cooling/R50_Alarms/rung:15	REQ-016	TC-F2S-P201
REV-010	Pending	Medium	Alarms/Timers	ControllerTag/CFG_CHW_DP_LowAlarm_SP	REQ-016	TC-F2S-P201
REV-011	Pending	Medium	Alarms/Timers	ControllerTag/CFG_CHW_DP_SP	REQ-013	TC-F2S-P201
REV-012	Pending	Medium	Alarms/Timers	ControllerTag/CFG_CHW_SuctionLowTrip_SP	REQ-016	TC-F2S-P201
REV-013	Pending	Medium	Sequences	ControllerTag/CFG_LeadRotation_hr	REQ-016	TC-F2S-P201
REV-014	Pending	Medium	Alarms/Timers	ControllerTag/CFG_PumpFlowProofTimeout_ms	REQ-016	TC-F2S-P201
REV-015	Pending	Medium	Sequences	ControllerTag/CFG_PumpPostRun_ms	REQ-013	TC-F2S-P201
REV-016	Pending	Medium	Alarms/Timers	ControllerTag/CFG_SAT_High_SP	REQ-016	TC-F2S-P201

REV-001 — CHWP02 Failover Request Logic Test

Status: Accepted | Severity: Medium | Class: Sequences

Classification: Regression

Location: P_DC_Cooling/R30_CHWP_Control/rung:2

Equipment: CHWP01

L5X Diff Evidence

Baseline: [XIC(SYS_RunRequest),XIC(SYS_PostRunActive)]XIC(SYS_LeadPump1Selected)XIC(DI_CHWP01_Fault)XIO(DI_CHWP02_Fault)OTE(CHWP02_FailoverRequest);

Revised: AFI()[XIC(SYS_RunRequest),XIC(SYS_PostRunActive)]XIC(SYS_LeadPump1Selected)XIC(DI_CHWP01_Fault)XIO(DI_CHWP02_Fault)OTE(CHWP02_FailoverRequest);

Mapped Requirement

MAP-DIFF-001 (pdf-p2-b1)

DC1 Cooling Sequence of Operations Rev B | SYNTHETIC TEST DATA ONLY. NOT FOR FIELD INSTALLATION, COMMISSIONING, OR OPERATIONAL USE. 1. Purpose and governing use This document defines the intended control behavior for the synthetic DC1 cooling project. Revision B is the governing requirement source for prototype testing. The baseline PLC export represents Revision A. The revised PLC export is intended to implement Revision B but contains deliberately planted defects. 2. System overview The system serves two data halls using two chilled-water pumps, one common isolation valve, and two air- handling units. One chilled-water pump is selected as lead and the other is standby. Both AHUs can operate when chilled-water flow is proven. Pump speed maintains common chilled-water differential pressure. Equipment Function Lead or standby chilled-water pump with VFD, run CHWP-01 feedback, fault, and flow proof. Lead or standby chilled-water pump with VFD, run CHWP-02 feedback, fault, and flow proof. Common chilled-water isolation valve with open CHWV-01 feedback. Data Hall 1 supply air unit with VFD, run feedback, AHU-01 fault, airflow proof, and supply-air temperature. Data Hall 2 supply air un

Regression Test Plan

Test ID: TC-F2S-P102

Purpose: To verify that the CHWP02 failover request logic operates correctly under the new AFI instruction implementation, ensuring proper pump operation during fault conditions.

Change: Modified rung 2 logic to use AFI instruction for CHWP02 failover request based on system run and pump selection conditions.

Equipment: CHWP01

Prerequisites: PLC is powered on and in test mode.; DI_CHWP01_Fault is set to 0 (no fault).; DI_CHWP02_Fault is set to 0 (no fault).; SYS_RunRequest is set to 1 (run request active).; SYS_PostRunActive is set to 1 (post run active).; SYS_LeadPump1Selected is set to 1 (pump 1 is selected).

Steps: 1. Set DI_CHWP01_Fault to 1 (simulate fault on CHWP01). | 2. Verify the state of CHWP02_FailoverRequest tag. | 3. Reset DI_CHWP01_Fault to 0 (clear fault on CHWP01). | 4. Set SYS_LeadPump2Selected to 1 (select pump 2 as lead). | 5. Set DI_CHWP02_Fault to 1 (simulate fault on CHWP02).

Expected result: CHWP02_FailoverRequest should be set to 1 when CHWP01 is in fault and remain active until CHWP02 also faults or conditions change.

REV-002 — CHWP01 Failover Request Logic Test

Status: Pending | Severity: Medium | Class: Sequences

Classification: Regression

Location: P_DC_Cooling/R30_CHWP_Control/rung:3

Equipment: CHWP02

L5X Diff Evidence

Baseline: [XIC(SYS_RunRequest),XIC(SYS_PostRunActive)]XIC(SYS_LeadPump2Selected)XIC(DI_CHWP02_Fault)XIO(DI_CHWP01_Fault)OTE(CHWP01_FailoverRequest);

Revised: AFI()[XIC(SYS_RunRequest),XIC(SYS_PostRunActive)]XIC(SYS_LeadPump2Selected)XIC(DI_CHWP02_Fault)XIO(DI_CHWP01_Fault)OTE(CHWP01_FailoverRequest);

Mapped Requirement

MAP-DIFF-002 (pdf-p2-b1)

DC1 Cooling Sequence of Operations Rev B | SYNTHETIC TEST DATA ONLY. NOT FOR FIELD INSTALLATION, COMMISSIONING, OR OPERATIONAL USE. 1. Purpose and governing use This document defines the intended control behavior for the synthetic DC1 cooling project. Revision B is the governing requirement source for prototype testing. The baseline PLC export represents Revision A. The revised PLC export is intended to implement Revision B but contains deliberately planted defects. 2. System overview The system serves two data halls using two chilled-water pumps, one common isolation valve, and two air- handling units. One chilled-water pump is selected as lead and the other is standby. Both AHUs can operate when chilled-water flow is proven. Pump speed maintains common chilled-water differential pressure. Equipment Function Lead or standby chilled-water pump with VFD, run CHWP-01 feedback, fault, and flow proof. Lead or standby chilled-water pump with VFD, run CHWP-02 feedback, fault, and flow proof. Common chilled-water isolation valve with open CHWV-01 feedback. Data Hall 1 supply air unit with VFD, run feedback, AHU-01 fault, airflow proof, and supply-air temperature. Data Hall 2 supply air un

Regression Test Plan

Test ID: TC-F2S-P201

Purpose: To verify that the CHWP01 failover request logic operates correctly with the new AFI() instruction under specified conditions.

Change: Modified rung 3 logic to use AFI() instruction for CHWP01 failover request based on system run and fault conditions.

Equipment: CHWP02

Prerequisites: PLC is powered on and in test mode.; DI_CHWP01_Fault is set to 0 (no fault).; DI_CHWP02_Fault is set to 0 (no fault).; SYS_RunRequest is set to 1 (system running).; SYS_PostRunActive is set to 1 (post run active).; SYS_LeadPump2Selected is set to 1 (pump 2 selected).

Steps: 1. Set DI_CHWP01_Fault to 0. | 2. Set DI_CHWP02_Fault to 0. | 3. Set SYS_RunRequest to 1. | 4. Set SYS_PostRunActive to 1. | 5. Set SYS_LeadPump2Selected to 1. | 6. Monitor CHWP01_FailoverRequest tag for changes.

Expected result: CHWP01_FailoverRequest is set to 1 when all prerequisites are met and no faults are present.

REV-003 — CHWP02 Permissive Interlock Test

Status: Pending | Severity: Critical | Class: Interlocks

Classification: Regression

Location: P_DC_Cooling/R30_CHWP_Control/rung:5

Equipment: CHWP02

L5X Diff Evidence

Baseline: XIC(DI_EPO_Healthy)XIO(DI_FireAlarm_Active)XIO(DI_LeakDetected)GRT(AI_CHW_Suction_PSI,CFG_CHW_SuctionLowTrip_SP)XIC(DI_CHW_IsolationValve_OpenFb)XIO(DI_CHWP02_Fault)OTE(CHWP02_Permissive);

Revised: XIC(DI_EPO_Healthy)XIO(DI_FireAlarm_Active)GRT(AI_CHW_Suction_PSI,CFG_CHW_SuctionLowTrip_SP)XIC(DI_CHW_IsolationValve_OpenFb)XIO(DI_CHWP02_Fault)OTE(CHWP02_Permissive);

Mapped Requirement

REQ-013 (pdf-p4-b1)

DC1 Cooling Sequence of Operations Rev B | SYNTHETIC TEST DATA ONLY. NOT FOR FIELD INSTALLATION, COMMISSIONING, OR OPERATIONAL USE. Requirement ID Requirement AHU automatic start shall require proven chilled- REQ-013 water flow. Each AHU shall prove airflow within 15 seconds of REQ-014 a start command or generate a fail-to-start alarm. Manual equipment commands shall remain subject to emergency power-off, fire, leak, REQ-015 equipment fault, and required process permissives. Each AHU supply-air high-temperature alarm shall REQ-016 occur above 88.0 deg F for 30 seconds. Each AHU supply-air high-high-temperature alarm REQ-017 shall occur at or above 95.0 deg F for 10 seconds. A low chilled-water differential-pressure alarm REQ-018 shall occur below 12.0 psi for 20 seconds while the cooling system is running. A data-hall high-temperature alarm shall occur REQ-019 above 90.0 deg F for 30 seconds. The chilled-water differential-pressure control REQ-020 setpoint shall be 20.0 psi. The active pump speed command shall be limited REQ-021 to 30 through 100 percent. AHU supply-fan speed shall be 55 percent from 78 REQ-022 through 82 deg F and 70 percent above 82 deg F. Alarm status shall be e

Regression Test Plan

Test ID: TC-F2S-P102

Purpose: To verify that the CHWP02 permissive interlock operates correctly without the leak detection condition, ensuring compliance with the updated control logic.

Change: Modified rung 5 logic to remove the leak detection condition from the CHWP02 permissive interlock.

Equipment: CHWP02

Prerequisites: PLC is powered and in RUN mode.; DI_EPO_Healthy is TRUE.; DI_FireAlarm_Active is FALSE.; AI_CHW_Suction_PSI is set above CFG_CHW_SuctionLowTrip_SP.; DI_CHW_IsolationValve_OpenFb is TRUE.; DI_CHWP02_Fault is FALSE.

Steps: 1. Set AI_CHW_Suction_PSI to a value above CFG_CHW_SuctionLowTrip_SP. | 2. Verify DI_EPO_Healthy is TRUE and DI_FireAlarm_Active is FALSE. | 3. Check DI_CHW_IsolationValve_OpenFb is TRUE. | 4. Check DI_CHWP02_Fault is FALSE. | 5. Monitor the state of CHWP02_Permissive.

Expected result: CHWP02_Permissive should be TRUE when all conditions are met, indicating the interlock is functioning correctly without the leak detection condition.

REV-004 — Test CHWP02 Flow Proof Timer Adjustment

Status: Unresolved | Severity: Medium | Class: Alarms/Timers

Classification: Behavioral change

Location: P_DC_Cooling/R30_CHWP_Control/rung:9

Equipment: CHWP02

L5X Diff Evidence

Baseline: `XIC(CMD_CHWP02_Start)XIO(DI_CHWP02_FlowProof)TON(TMR_CHWP02_FlowProof,CFG_PumpFlowProofTimeout_ms,0);`

Revised: `XIC(CMD_CHWP02_Start)XIO(DI_CHWP02_FlowProof)TON(TMR_CHWP02_FlowProof,25000,0);`

Mapped Requirement

MAP-DIFF-004 (pdf-p2-b1)

DC1 Cooling Sequence of Operations Rev B | SYNTHETIC TEST DATA ONLY. NOT FOR FIELD INSTALLATION, COMMISSIONING, OR OPERATIONAL USE. 1. Purpose and governing use This document defines the intended control behavior for the synthetic DC1 cooling project. Revision B is the governing requirement source for prototype testing. The baseline PLC export represents Revision A. The revised PLC export is intended to implement Revision B but contains deliberately planted defects. 2. System overview The system serves two data halls using two chilled-water pumps, one common isolation valve, and two air- handling units. One chilled-water pump is selected as lead and the other is standby. Both AHUs can operate when chilled-water flow is proven. Pump speed maintains common chilled-water differential pressure. Equipment Function Lead or standby chilled-water pump with VFD, run CHWP-01 feedback, fault, and flow proof. Lead or standby chilled-water pump with VFD, run CHWP-02 feedback, fault, and flow proof. Common chilled-water isolation valve with open CHWV-01 feedback. Data Hall 1 supply air unit with VFD, run feedback, AHU-01 fault, airflow proof, and supply-air temperature. Data Hall 2 supply air un

Regression Test Plan

Test ID: TC-F2S-P201

Purpose: To verify that the flow proof timer for CHWP02 operates correctly with the new fixed timeout value of 25000 ms.

Change: Updated TMR_CHWP02_FlowProof timer from CFG_PumpFlowProofTimeout_ms to a fixed value of 25000 ms.

Equipment: CHWP02

Prerequisites: PLC is powered on and in test mode.; CMD_CHWP02_Start is set to TRUE.; DI_CHWP02_FlowProof is set to FALSE.

Steps: 1. Set CMD_CHWP02_Start to TRUE. | 2. Set DI_CHWP02_FlowProof to FALSE. | 3. Wait for 25000 ms. | 4. Check the state of TMR_CHWP02_FlowProof after 25000 ms.

Expected result: TMR_CHWP02_FlowProof should complete after 25000 ms, indicating the timer functions correctly with the new setting.

REV-005 — Test AHU02 Permissive Logic

Status: Pending | Severity: Critical | Class: Interlocks

Classification: Regression

Location: P_DC_Cooling/R40_AHU_Control/rung:3

Equipment: AHU02

L5X Diff Evidence

Baseline: XIC(DI_EPO_Healthy)XIO(DI_FireAlarm_Active)XIO(DI_LeakDetected)XIC(SYS_PumpReady)XIO(DI_AHU02_Fault)OTE(AHU02_Permissive);

Revised:

XIC(DI_EPO_Healthy)XIO(DI_LeakDetected)XIC(SYS_PumpReady)XIO(DI_AHU02_Fault)OTE(AHU02_Permissive);

Mapped Requirement

REQ-001 (pdf-p3-b1)

DC1 Cooling Sequence of Operations Rev B | SYNTHETIC TEST DATA ONLY. NOT FOR FIELD INSTALLATION, COMMISSIONING, OR OPERATIONAL USE. 5. Normal control The active chilled-water pump speed command shall target 20.0 psi differential pressure and shall be clamped between 30 and 100 percent. AHU supply-fan speed shall be 55 percent from 78 through 82 deg F and 70 percent above 82 deg F. Fan speed shall be zero when the AHU is not commanded to run. 6. Shutdown and protective behavior Normal loss of cooling demand stops the AHUs and initiates a 120-second pump and valve post-run. Emergency power-off, leak, and equipment trips immediately remove the affected equipment command. Fire alarm immediately stops both AHUs. Manual mode does not bypass these protections. 7. Detailed requirements Requirement ID Requirement The cooling system shall operate automatically REQ-001 only when HMI_System_AutoEnable and DI_Remote_Enable are both true. All equipment start commands shall be blocked REQ-002 when the emergency power-off loop is not healthy. A fire alarm shall immediately stop both AHU REQ-003 supply fans. Manual commands shall not bypass this interlock. A detected chilled-water leak shall stop b

Regression Test Plan

Test ID: TC-F2S-P201

Purpose: To verify that the updated permissive logic for AHU02 operates correctly without the fire alarm condition affecting the permissive state.

Change: Removed XIO(DI_FireAlarm_Active) from the permissive logic for AHU02.

Equipment: AHU02

Prerequisites: PLC is powered on and in a test mode.; DI_EPO_Healthy is set to true.; SYS_PumpReady is set to true.; DI_AHU02_Fault is set to false.

Steps: 1. Set DI_LeakDetected to false. | 2. Set DI_FireAlarm_Active to false. | 3. Verify AHU02_Permissive is true. | 4. Set DI_AHU02_Fault to true. | 5. Verify AHU02_Permissive is false. | 6. Set DI_AHU02_Fault to false. | 7. Set DI_LeakDetected to true. | 8. Verify AHU02_Permissive is false.

Expected result: AHU02_Permissive should be true when DI_EPO_Healthy is true, DI_FireAlarm_Active is false, DI_LeakDetected is false, and DI_AHU02_Fault is false. AHU02_Permissive should be false when DI_AHU02_Fault is true or DI_LeakDetected is true.

REV-006 — AHU02 Control Mode Verification

Status: Pending | Severity: High | Class: Modes

Classification: Behavioral change

Location: P_DC_Cooling/R40_AHU_Control/rung:5

Equipment: AHU02

L5X Diff Evidence

Baseline: [XIC(AHU02_AutoRequest),XIC(HMI_AHU02_HandCmd)]XIC(AHU02_Permissive)OTE(CMD_AHU02_Start);

Revised: [XIC(AHU02_AutoRequest),XIC(HMI_AHU01_HandCmd)]XIC(AHU02_Permissive)OTE(CMD_AHU02_Start);

Mapped Requirement

MAP-DIFF-006 (pdf-p2-b1)

DC1 Cooling Sequence of Operations Rev B | SYNTHETIC TEST DATA ONLY. NOT FOR FIELD INSTALLATION, COMMISSIONING, OR OPERATIONAL USE. 1. Purpose and governing use This document defines the intended control behavior for the synthetic DC1 cooling project. Revision B is the governing requirement source for prototype testing. The baseline PLC export represents Revision A. The revised PLC export is intended to implement Revision B but contains deliberately planted defects. 2. System overview The system serves two data halls using two chilled-water pumps, one common isolation valve, and two air- handling units. One chilled-water pump is selected as lead and the other is standby. Both AHUs can operate when chilled-water flow is proven. Pump speed maintains common chilled-water differential pressure. Equipment Function Lead or standby chilled-water pump with VFD, run CHWP-01 feedback, fault, and flow proof. Lead or standby chilled-water pump with VFD, run CHWP-02 feedback, fault, and flow proof. Common chilled-water isolation valve with open CHWV-01 feedback. Data Hall 1 supply air unit with VFD, run feedback, AHU-01 fault, airflow proof, and supply-air temperature. Data Hall 2 supply air un

Regression Test Plan

Test ID: TC-F2S-P201

Purpose: To verify that the control logic for AHU02 correctly responds to the new hand command input from HMI_AHU01 while maintaining permissive conditions.

Change: Modified rung 5 to change the hand command input from HMI_AHU02_HandCmd to HMI_AHU01_HandCmd for AHU02 control.

Equipment: AHU02

Prerequisites: PLC is powered on and in test mode.; HMI_AHU01_HandCmd is simulated as active (true).; AHU02_Permissive is simulated as active (true).

Steps: 1. Simulate HMI_AHU01_HandCmd as true. | 2. Simulate AHU02_Permissive as true. | 3. Monitor CMD_AHU02_Start tag for state change. | 4. Simulate HMI_AHU01_HandCmd as false. | 5. Monitor CMD_AHU02_Start tag for state change.

Expected result: CMD_AHU02_Start should be true when HMI_AHU01_HandCmd and AHU02_Permissive are true, and false when HMI_AHU01_HandCmd is false.

REV-007 — Test AHU01 Airflow Proof Timer Adjustment

Status: Pending | Severity: Medium | Class: Alarms/Timers

Classification: Behavioral change

Location: P_DC_Cooling/R40_AHU_Control/rung:6

Equipment: AHU01

L5X Diff Evidence

Baseline: `XIC(CMD_AHU01_Start)XIO(DI_AHU01_AirflowProof)TON(TMR_AHU01_AirflowProof,CFG_AHU_AirflowProofTimeout_ms,0);`

Revised: `XIC(CMD_AHU01_Start)XIO(DI_AHU01_AirflowProof)TON(TMR_AHU01_AirflowProof,5000,0);`

Mapped Requirement

REQ-013 (pdf-p4-b1)

DC1 Cooling Sequence of Operations Rev B | SYNTHETIC TEST DATA ONLY. NOT FOR FIELD INSTALLATION, COMMISSIONING, OR OPERATIONAL USE. Requirement ID Requirement AHU automatic start shall require proven chilled- REQ-013 water flow. Each AHU shall prove airflow within 15 seconds of REQ-014 a start command or generate a fail-to-start alarm. Manual equipment commands shall remain subject to emergency power-off, fire, leak, REQ-015 equipment fault, and required process permissives. Each AHU supply-air high-temperature alarm shall REQ-016 occur above 88.0 deg F for 30 seconds. Each AHU supply-air high-high-temperature alarm REQ-017 shall occur at or above 95.0 deg F for 10 seconds. A low chilled-water differential-pressure alarm REQ-018 shall occur below 12.0 psi for 20 seconds while the cooling system is running. A data-hall high-temperature alarm shall occur REQ-019 above 90.0 deg F for 30 seconds. The chilled-water differential-pressure control REQ-020 setpoint shall be 20.0 psi. The active pump speed command shall be limited REQ-021 to 30 through 100 percent. AHU supply-fan speed shall be 55 percent from 78 REQ-022 through 82 deg F and 70 percent above 82 deg F. Alarm status shall be e

Regression Test Plan

Test ID: TC-F2S-P101

Purpose: To verify that the AHU01 airflow proof timer correctly enforces a 5000 ms timeout for airflow proof validation upon receiving a start command.

Change: Adjusted the airflow proof timer from a configurable timeout to a fixed timeout of 5000 ms.

Equipment: AHU01

Prerequisites: PLC is powered on and in a test mode.; AHU01 is connected and operational in the test environment.; DI_AHU01_AirflowProof is set to '0' (not proven).

Steps: 1. Set CMD_AHU01_Start to '1' to initiate the AHU01 start command. | 2. Wait for 5000 ms after setting CMD_AHU01_Start. | 3. Check the state of TMR_AHU01_AirflowProof after the wait period. | 4. Set DI_AHU01_AirflowProof to '1' (proven). | 5. Set CMD_AHU01_Start to '0' to stop the command.

Expected result: TMR_AHU01_AirflowProof should time out after 5000 ms and transition to the 'done' state if DI_AHU01_AirflowProof remains '0'. If DI_AHU01_AirflowProof is set to '1' before the timeout, the timer should reset and not time out.

REV-008 — Test TMR_AHU01_SAT_HighHigh Timer Configuration

Status: Pending | Severity: Medium | Class: Alarms/Timers

Classification: Behavioral change

Location: P_DC_Cooling/R50_Alarms/rung:13

Equipment: AHU01

L5X Diff Evidence

Baseline: GEQ(AI_AHU01_SAT_F,CFG_SAT_HighHigh_SP)TON(TMR_AHU01_SAT_HighHigh,CFG_SAT_HighHigh_Delay_ms,0);

Revised: GEQ(AI_AHU01_SAT_F,CFG_SAT_HighHigh_SP)TON(TMR_AHU01_SAT_HighHigh,60000,0);

Mapped Requirement

REQ-016 (pdf-p5-b1)

DC1 Cooling Sequence of Operations Rev B | SYNTHETIC TEST DATA ONLY. NOT FOR FIELD INSTALLATION, COMMISSIONING, OR OPERATIONAL USE. PLC Tag Description Value Units Requirement Supply-air high CFG_SAT_High_SP 88.0 deg F REQ-016 alarm threshold CFG_SAT_HighHig Supply-air high- 95.0 deg F REQ-017 h_SP high threshold CFG_SAT_High_Del Supply-air high 30000 ms REQ-016 ay_ms alarm delay CFG_SAT_HighHig Supply-air high- 10000 ms REQ-017 h_Delay_ms high alarm delay CHW differential- CFG_CHW_DP_Lo pressure low 12.0 psi REQ-018 wAlarm_SP alarm CFG_CHW_DP_Lo CHW DP low delay 20000 ms REQ-018 wAlarm_Delay_ms CFG_LeadRotation Lead rotation 168.0 hr REQ-012 _hr advisory interval CHW return- CFG_CHW_Return temperature-high 72.0 deg F REQ-026 High_SP threshold 9. Alarm behavior Alarm definitions, priorities, delays, messages, and operator responses are governed by the separate Alarm List Revision B. Where the PLC export conflicts with the Alarm List or this Sequence of Operations, the approved requirement and Alarm List shall govern for this synthetic test package. 10. Revision history Revision Date Change Initial synthetic design. CHW DP setpoint 18.0 psi, pump proof A 2026-07-10 timeout 10 second

Regression Test Plan

Test ID: TC-F2S-P201

Purpose: To verify that the timer for the high-high saturation alarm operates correctly with the new fixed delay setting.

Change: Modified timer configuration for high-high saturation alarm from a variable delay to a fixed delay of 60000 ms.

Equipment: AHU01

Prerequisites: PLC is powered on and in test mode.; AHU01 is connected and operational in the test environment.; CFG_SAT_HighHigh_SP is set to 88.0 deg F for testing.

Steps: 1. Set AI_AHU01_SAT_F to 95.0 deg F to trigger the high-high alarm condition. | 2. Wait for 60000 ms after the alarm condition is triggered. | 3. Check the status of the alarm system for the high-high saturation alarm.

Expected result: TMR_AHU01_SAT_HighHigh should activate after 60000 ms when AI_AHU01_SAT_F is at or above 95.0 deg F, indicating the alarm condition is functioning as intended.

REV-009 — Test AHU02 High High SAT Alarm Timer

Status: Pending | Severity: Medium | Class: Alarms/Timers

Classification: Behavioral change

Location: P_DC_Cooling/R50_Alarms/rung:15

Equipment: AHU02

L5X Diff Evidence

Baseline: GEQ(AI_AHU02_SAT_F,CFG_SAT_HighHigh_SP)TON(TMR_AHU02_SAT_HighHigh,CFG_SAT_HighHigh_Delay_ms,0);

Revised: GEQ(AI_AHU02_SAT_F,CFG_SAT_HighHigh_SP)TON(TMR_AHU02_SAT_HighHigh,60000,0);

Mapped Requirement

REQ-016 (pdf-p5-b1)

DC1 Cooling Sequence of Operations Rev B | SYNTHETIC TEST DATA ONLY. NOT FOR FIELD INSTALLATION, COMMISSIONING, OR OPERATIONAL USE. PLC Tag Description Value Units Requirement Supply-air high CFG_SAT_High_SP 88.0 deg F REQ-016 alarm threshold CFG_SAT_HighHig Supply-air high- 95.0 deg F REQ-017 h_SP high threshold CFG_SAT_High_Del Supply-air high 30000 ms REQ-016 ay_ms alarm delay CFG_SAT_HighHig Supply-air high- 10000 ms REQ-017 h_Delay_ms high alarm delay CHW differential- CFG_CHW_DP_Lo pressure low 12.0 psi REQ-018 wAlarm_SP alarm CFG_CHW_DP_Lo CHW DP low delay 20000 ms REQ-018 wAlarm_Delay_ms CFG_LeadRotation Lead rotation 168.0 hr REQ-012 _hr advisory interval CHW return- CFG_CHW_Return temperature-high 72.0 deg F REQ-026 High_SP threshold 9. Alarm behavior Alarm definitions, priorities, delays, messages, and operator responses are governed by the separate Alarm List Revision B. Where the PLC export conflicts with the Alarm List or this Sequence of Operations, the approved requirement and Alarm List shall govern for this synthetic test package. 10. Revision history Revision Date Change Initial synthetic design. CHW DP setpoint 18.0 psi, pump proof A 2026-07-10 timeout 10 second

Regression Test Plan

Test ID: TC-F2S-P201

Purpose: To verify that the high high SAT alarm timer functions correctly with the new fixed delay setting.

Change: Updated the timer for the high high SAT alarm from a configurable delay to a fixed delay of 60000 ms.

Equipment: AHU02

Prerequisites: PLC is powered on and in run mode.; AHU02 is connected and operational in the test environment.

Steps: 1. Simulate the supply air temperature (AI_AHU02_SAT_F) to exceed the high high threshold (CFG_SAT_HighHigh_SP = 95.0 deg F). | 2. Check the status of TMR_AHU02_SAT_HighHigh after 60000 ms. | 3. Verify that the high high SAT alarm is triggered after the timer completes.

Expected result: The high high SAT alarm should activate after 60000 ms when the supply air temperature exceeds 95.0 deg F, confirming the timer operates as intended.

REV-010 — Verify CFG_CHW_DP_LowAlarm_SP Change to 8 psi

Status: Pending | Severity: Medium | Class: Alarms/Timers

Classification: Behavioral change

Location: ControllerTag/CFG_CHW_DP_LowAlarm_SP

Tag: CFG_CHW_DP_LowAlarm_SP

L5X Diff Evidence

Baseline: CFG_CHW_DP_LowAlarm_SP = 12

Revised: CFG_CHW_DP_LowAlarm_SP = 8

Mapped Requirement

REQ-016 (pdf-p5-b1)

DC1 Cooling Sequence of Operations Rev B | SYNTHETIC TEST DATA ONLY. NOT FOR FIELD INSTALLATION, COMMISSIONING, OR OPERATIONAL USE. PLC Tag Description Value Units Requirement Supply-air high CFG_SAT_High_SP 88.0 deg F REQ-016 alarm threshold CFG_SAT_HighHig Supply-air high- 95.0 deg F REQ-017 h_SP high threshold CFG_SAT_High_Del Supply-air high 30000 ms REQ-016 ay_ms alarm delay CFG_SAT_HighHig Supply-air high- 10000 ms REQ-017 h_Delay_ms high alarm delay CHW differential- CFG_CHW_DP_Lo pressure low 12.0 psi REQ-018 wAlarm_SP alarm CFG_CHW_DP_Lo CHW DP low delay 20000 ms REQ-018 wAlarm_Delay_ms CFG_LeadRotation Lead rotation 168.0 hr REQ-012 _hr advisory interval CHW return- CFG_CHW_Return temperature-high 72.0 deg F REQ-026 High_SP threshold 9. Alarm behavior Alarm definitions, priorities, delays, messages, and operator responses are governed by the separate Alarm List Revision B. Where the PLC export conflicts with the Alarm List or this Sequence of Operations, the approved requirement and Alarm List shall govern for this synthetic test package. 10. Revision history Revision Date Change Initial synthetic design. CHW DP setpoint 18.0 psi, pump proof A 2026-07-10 timeout 10 second

Regression Test Plan

Test ID: TC-F2S-P201

Purpose: To ensure that the low alarm setpoint for CHW differential pressure is correctly set to 8 psi and that the alarm triggers appropriately.

Change: Changed CFG_CHW_DP_LowAlarm_SP from 12 psi to 8 psi.

Equipment: CHW System

Prerequisites: PLC is powered on and in run mode.; CHW system is connected and operational in the test environment.

Steps: 1. Set CFG_CHW_DP_LowAlarm_SP to 8 psi in the PLC configuration. | 2. Simulate a CHW differential pressure of 7 psi using AI_CHW_DP_PSI. | 3. Check the status of ALM_CHW_DP_Low. | 4. Reset the alarm and return the CHW differential pressure to normal (above 8 psi).

Expected result: The alarm for low CHW differential pressure activates at 7 psi and resets correctly when pressure is above 8 psi.

REV-011 — Chilled Water Differential Pressure Setpoint Verification

Status: Pending | Severity: Medium | Class: Alarms/Timers

Classification: Behavioral change

Location: ControllerTag/CFG_CHW_DP_SP

Tag: CFG_CHW_DP_SP

L5X Diff Evidence

Baseline: CFG_CHW_DP_SP = 18

Revised: CFG_CHW_DP_SP = 20

Mapped Requirement

REQ-013 (pdf-p4-b1)

DC1 Cooling Sequence of Operations Rev B | SYNTHETIC TEST DATA ONLY. NOT FOR FIELD INSTALLATION, COMMISSIONING, OR OPERATIONAL USE. Requirement ID Requirement AHU automatic start shall require proven chilled- REQ-013 water flow. Each AHU shall prove airflow within 15 seconds of REQ-014 a start command or generate a fail-to-start alarm. Manual equipment commands shall remain subject to emergency power-off, fire, leak, REQ-015 equipment fault, and required process permissives. Each AHU supply-air high-temperature alarm shall REQ-016 occur above 88.0 deg F for 30 seconds. Each AHU supply-air high-high-temperature alarm REQ-017 shall occur at or above 95.0 deg F for 10 seconds. A low chilled-water differential-pressure alarm REQ-018 shall occur below 12.0 psi for 20 seconds while the cooling system is running. A data-hall high-temperature alarm shall occur REQ-019 above 90.0 deg F for 30 seconds. The chilled-water differential-pressure control REQ-020 setpoint shall be 20.0 psi. The active pump speed command shall be limited REQ-021 to 30 through 100 percent. AHU supply-fan speed shall be 55 percent from 78 REQ-022 through 82 deg F and 70 percent above 82 deg F. Alarm status shall be e

Regression Test Plan

Test ID: TC-F2S-P201

Purpose: To verify that the chilled-water differential pressure setpoint is correctly set to 20 psi and that the low alarm functionality operates as intended.

Change: Updated CFG_CHW_DP_SP from 18 to 20 psi to meet operational requirements.

Equipment: AHU-01

Prerequisites: PLC is powered on and in test mode.; AHU-01 is connected to the test bench with appropriate sensors and alarms configured.

Steps: 1. Set CFG_CHW_DP_SP to 20 psi in the PLC configuration. | 2. Simulate a condition where the chilled-water differential pressure drops below 12 psi. | 3. Wait for 20 seconds to observe if the low chilled-water differential-pressure alarm activates. | 4. Reset the alarm and return CFG_CHW_DP_SP to 20 psi.

Expected result: The low chilled-water differential-pressure alarm activates within 20 seconds when the pressure drops below 12 psi, confirming the setpoint change is effective and alarms function correctly.

REV-012 — Test CFG_CHW_SuctionLowTrip_SP Adjustment

Status: Pending | Severity: Medium | Class: Alarms/Timers

Classification: Behavioral change

Location: ControllerTag/CFG_CHW_SuctionLowTrip_SP

Tag: CFG_CHW_SuctionLowTrip_SP

L5X Diff Evidence

Baseline: CFG_CHW_SuctionLowTrip_SP = 5

Revised: CFG_CHW_SuctionLowTrip_SP = 2

Mapped Requirement

REQ-016 (pdf-p5-b1)

DC1 Cooling Sequence of Operations Rev B | SYNTHETIC TEST DATA ONLY. NOT FOR FIELD INSTALLATION, COMMISSIONING, OR OPERATIONAL USE. PLC Tag Description Value Units Requirement Supply-air high CFG_SAT_High_SP 88.0 deg F REQ-016 alarm threshold CFG_SAT_HighHig Supply-air high- 95.0 deg F REQ-017 h_SP high threshold CFG_SAT_High_Del Supply-air high 30000 ms REQ-016 ay_ms alarm delay CFG_SAT_HighHig Supply-air high- 10000 ms REQ-017 h_Delay_ms high alarm delay CHW differential- CFG_CHW_DP_Lo pressure low 12.0 psi REQ-018 wAlarm_SP alarm CFG_CHW_DP_Lo CHW DP low delay 20000 ms REQ-018 wAlarm_Delay_ms CFG_LeadRotation Lead rotation 168.0 hr REQ-012 _hr advisory interval CHW return- CFG_CHW_Return temperature-high 72.0 deg F REQ-026 High_SP threshold 9. Alarm behavior Alarm definitions, priorities, delays, messages, and operator responses are governed by the separate Alarm List Revision B. Where the PLC export conflicts with the Alarm List or this Sequence of Operations, the approved requirement and Alarm List shall govern for this synthetic test package. 10. Revision history Revision Date Change Initial synthetic design. CHW DP setpoint 18.0 psi, pump proof A 2026-07-10 timeout 10 second

Regression Test Plan

Test ID: TC-F2S-P201

Purpose: To verify that the low trip setpoint for the CHW suction pressure is correctly set and that the alarm triggers as expected when the suction pressure falls below the new threshold.

Change: The CFG_CHW_SuctionLowTrip_SP tag was changed from 5 to 2.

Equipment: CHW Pump System

Prerequisites: PLC is powered on and in RUN mode.; CHW Pump System is connected and operational in the test environment.

Steps: 1. Set the CFG_CHW_SuctionLowTrip_SP to 2 in the PLC configuration. | 2. Simulate a suction pressure drop below 2 psi using AI_DH02_Temp_F. | 3. Check the status of CFG_CHW_DP_LowAlarm_SP after the pressure drop. | 4. Reset the alarm and restore suction pressure above 2 psi.

Expected result: The alarm CFG_CHW_DP_LowAlarm_SP should trigger when the suction pressure drops below 2 psi and reset when the pressure is restored above this threshold.

REV-013 — Verify CFG_LeadRotation_hr Change

Status: Pending | Severity: Medium | Class: Sequences

Classification: Behavioral change

Location: ControllerTag/CFG_LeadRotation_hr

Tag: CFG_LeadRotation_hr

L5X Diff Evidence

Baseline: CFG_LeadRotation_hr = 168

Revised: CFG_LeadRotation_hr = 24

Mapped Requirement

REQ-016 (pdf-p5-b1)

DC1 Cooling Sequence of Operations Rev B | SYNTHETIC TEST DATA ONLY. NOT FOR FIELD INSTALLATION, COMMISSIONING, OR OPERATIONAL USE. PLC Tag Description Value Units Requirement Supply-air high CFG_SAT_High_SP 88.0 deg F REQ-016 alarm threshold CFG_SAT_HighHig Supply-air high- 95.0 deg F REQ-017 h_SP high threshold CFG_SAT_High_Del Supply-air high 30000 ms REQ-016 ay_ms alarm delay CFG_SAT_HighHig Supply-air high- 10000 ms REQ-017 h_Delay_ms high alarm delay CHW differential- CFG_CHW_DP_Lo pressure low 12.0 psi REQ-018 wAlarm_SP alarm CFG_CHW_DP_Lo CHW DP low delay 20000 ms REQ-018 wAlarm_Delay_ms CFG_LeadRotation Lead rotation 168.0 hr REQ-012 _hr advisory interval CHW return- CFG_CHW_Return temperature-high 72.0 deg F REQ-026 High_SP threshold 9. Alarm behavior Alarm definitions, priorities, delays, messages, and operator responses are governed by the separate Alarm List Revision B. Where the PLC export conflicts with the Alarm List or this Sequence of Operations, the approved requirement and Alarm List shall govern for this synthetic test package. 10. Revision history Revision Date Change Initial synthetic design. CHW DP setpoint 18.0 psi, pump proof A 2026-07-10 timeout 10 second

Regression Test Plan

Test ID: TC-F2S-P201

Purpose: To ensure that the lead rotation interval is correctly set to 24 hours and that the system behaves as expected with this new configuration.

Change: Changed CFG_LeadRotation_hr from 168 hours to 24 hours to adjust the lead rotation interval for the cooling system.

Equipment: Cooling System Controller

Prerequisites: Cooling System Controller is powered on and operational.; PLC program is uploaded with the latest changes.

Steps: 1. Read the value of CFG_LeadRotation_hr from the PLC. | 2. Monitor the ALM_LeadRotationDue tag for alarm activation. | 3. Wait for 24 hours in simulation time and check the ALM_LeadRotationDue tag again.

Expected result: CFG_LeadRotation_hr is set to 24 hours, no alarm is triggered immediately, and the ALM_LeadRotationDue alarm activates after 24 hours as expected.

REV-014 — Verify CFG_PumpFlowProofTimeout_ms Change

Status: Pending | Severity: Medium | Class: Alarms/Timers

Classification: Behavioral change

Location: ControllerTag/CFG_PumpFlowProofTimeout_ms

Tag: CFG_PumpFlowProofTimeout_ms

L5X Diff Evidence

Baseline: CFG_PumpFlowProofTimeout_ms = 10000

Revised: CFG_PumpFlowProofTimeout_ms = 15000

Mapped Requirement

REQ-016 (pdf-p5-b1)

DC1 Cooling Sequence of Operations Rev B | SYNTHETIC TEST DATA ONLY. NOT FOR FIELD INSTALLATION, COMMISSIONING, OR OPERATIONAL USE. PLC Tag Description Value Units Requirement Supply-air high CFG_SAT_High_SP 88.0 deg F REQ-016 alarm threshold CFG_SAT_HighHig Supply-air high- 95.0 deg F REQ-017 h_SP high threshold CFG_SAT_High_Del Supply-air high 30000 ms REQ-016 ay_ms alarm delay CFG_SAT_HighHig Supply-air high- 10000 ms REQ-017 h_Delay_ms high alarm delay CHW differential- CFG_CHW_DP_Lo pressure low 12.0 psi REQ-018 wAlarm_SP alarm CFG_CHW_DP_Lo CHW DP low delay 20000 ms REQ-018 wAlarm_Delay_ms CFG_LeadRotation Lead rotation 168.0 hr REQ-012 _hr advisory interval CHW return- CFG_CHW_Return temperature-high 72.0 deg F REQ-026 High_SP threshold 9. Alarm behavior Alarm definitions, priorities, delays, messages, and operator responses are governed by the separate Alarm List Revision B. Where the PLC export conflicts with the Alarm List or this Sequence of Operations, the approved requirement and Alarm List shall govern for this synthetic test package. 10. Revision history Revision Date Change Initial synthetic design. CHW DP setpoint 18.0 psi, pump proof A 2026-07-10 timeout 10 second

Regression Test Plan

Test ID: TC-F2S-P201

Purpose: To ensure the updated pump flow proof timeout is functioning correctly and meets the specified requirement for alarm delays.

Change: Updated CFG_PumpFlowProofTimeout_ms from 10000 ms to 15000 ms to extend the pump flow proof timeout duration.

Equipment: PUMP101

Prerequisites: PLC is powered on and in a test mode.; All related tags are accessible and configured in the PLC.

Steps: 1. Set CFG_PumpFlowProofTimeout_ms to 15000 ms in the PLC configuration. | 2. Trigger a pump flow proof condition by simulating a flow failure. | 3. Check if the alarm for pump flow proof timeout activates after 15000 ms. | 4. Reset the alarm and check if the system returns to normal operation.

Expected result: The pump flow proof timeout alarm should activate after 15000 ms and reset correctly without any errors.

REV-015 — Test CFG_PumpPostRun_ms Change

Status: Pending | Severity: Medium | Class: Sequences

Classification: Behavioral change

Location: ControllerTag/CFG_PumpPostRun_ms

Tag: CFG_PumpPostRun_ms

L5X Diff Evidence

Baseline: CFG_PumpPostRun_ms = 120000

Revised: CFG_PumpPostRun_ms = 0

Mapped Requirement

REQ-013 (pdf-p4-b1)

DC1 Cooling Sequence of Operations Rev B | SYNTHETIC TEST DATA ONLY. NOT FOR FIELD INSTALLATION, COMMISSIONING, OR OPERATIONAL USE. Requirement ID Requirement AHU automatic start shall require proven chilled- REQ-013 water flow. Each AHU shall prove airflow within 15 seconds of REQ-014 a start command or generate a fail-to-start alarm. Manual equipment commands shall remain subject to emergency power-off, fire, leak, REQ-015 equipment fault, and required process permissives. Each AHU supply-air high-temperature alarm shall REQ-016 occur above 88.0 deg F for 30 seconds. Each AHU supply-air high-high-temperature alarm REQ-017 shall occur at or above 95.0 deg F for 10 seconds. A low chilled-water differential-pressure alarm REQ-018 shall occur below 12.0 psi for 20 seconds while the cooling system is running. A data-hall high-temperature alarm shall occur REQ-019 above 90.0 deg F for 30 seconds. The chilled-water differential-pressure control REQ-020 setpoint shall be 20.0 psi. The active pump speed command shall be limited REQ-021 to 30 through 100 percent. AHU supply-fan speed shall be 55 percent from 78 REQ-022 through 82 deg F and 70 percent above 82 deg F. Alarm status shall be e

Regression Test Plan

Test ID: TC-F2S-P201

Purpose: To verify that the pump operates correctly without a post-run delay, ensuring compliance with the cooling sequence requirements.

Change: Changed CFG_PumpPostRun_ms from 120000 ms to 0 ms, affecting the post-run delay for the pump.

Equipment: AHU-01, PUMP-01

Prerequisites: PLC is powered on and in a test mode.; AHU-01 and PUMP-01 are connected and operational in the test environment.

Steps: 1. Set CFG_PumpPostRun_ms to 0 ms in the PLC configuration. | 2. Command AHU-01 to start. | 3. Check if airflow is proven within 15 seconds after the start command. | 4. Verify that no fail-to-start alarm is generated during the test.

Expected result: AHU-01 starts successfully, airflow is proven within 15 seconds, and no fail-to-start alarm is triggered.

REV-016 — Verify CFG_SAT_High_SP Change to 88°F

Status: Pending | Severity: Medium | Class: Alarms/Timers

Classification: Behavioral change

Location: ControllerTag/CFG_SAT_High_SP

Tag: CFG_SAT_High_SP

L5X Diff Evidence

Baseline: CFG_SAT_High_SP = 85

Revised: CFG_SAT_High_SP = 88

Mapped Requirement

REQ-016 (pdf-p5-b1)

DC1 Cooling Sequence of Operations Rev B | SYNTHETIC TEST DATA ONLY. NOT FOR FIELD INSTALLATION, COMMISSIONING, OR OPERATIONAL USE. PLC Tag Description Value Units Requirement Supply-air high CFG_SAT_High_SP 88.0 deg F REQ-016 alarm threshold CFG_SAT_HighHig Supply-air high- 95.0 deg F REQ-017 h_SP high threshold CFG_SAT_High_Del Supply-air high 30000 ms REQ-016 ay_ms alarm delay CFG_SAT_HighHig Supply-air high- 10000 ms REQ-017 h_Delay_ms high alarm delay CHW differential- CFG_CHW_DP_Lo pressure low 12.0 psi REQ-018 wAlarm_SP alarm CFG_CHW_DP_Lo CHW DP low delay 20000 ms REQ-018 wAlarm_Delay_ms CFG_LeadRotation Lead rotation 168.0 hr REQ-012 _hr advisory interval CHW return- CFG_CHW_Return temperature-high 72.0 deg F REQ-026 High_SP threshold 9. Alarm behavior Alarm definitions, priorities, delays, messages, and operator responses are governed by the separate Alarm List Revision B. Where the PLC export conflicts with the Alarm List or this Sequence of Operations, the approved requirement and Alarm List shall govern for this synthetic test package. 10. Revision history Revision Date Change Initial synthetic design. CHW DP setpoint 18.0 psi, pump proof A 2026-07-10 timeout 10 second

Regression Test Plan

Test ID: TC-F2S-P201

Purpose: To ensure that the updated high setpoint for supply-air temperature is correctly implemented and triggers the appropriate alarms.

Change: Updated CFG_SAT_High_SP from 85°F to 88°F to align with revised operational requirements.

Equipment: Cooling System Controller

Prerequisites: Cooling System Controller is powered on and operational.; PLC program is loaded with the latest revision containing the CFG_SAT_High_SP change.

Steps: 1. Set CFG_SAT_High_SP to 88°F in the PLC configuration. | 2. Simulate a supply-air temperature of 89°F. | 3. Check the alarm status for CFG_SAT_HighHigh_SP. | 4. Reset the alarm and simulate a supply-air temperature of 87°F. | 5. Check the alarm status for CFG_SAT_HighHigh_SP again.

Expected result: The alarm for CFG_SAT_HighHigh_SP should trigger when the supply-air temperature exceeds 88°F and should not trigger when below 88°F.